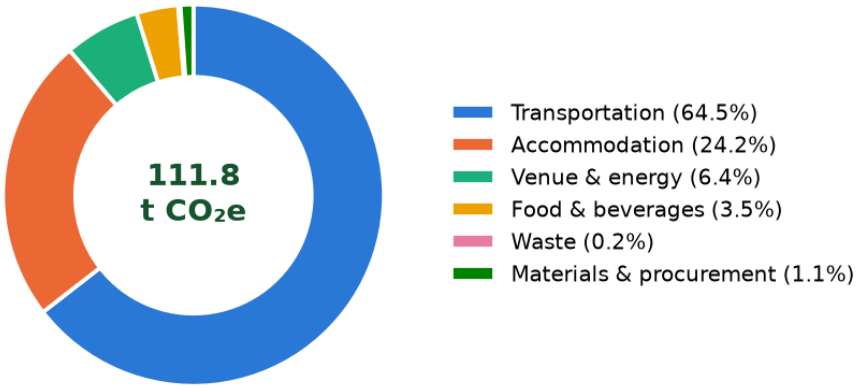
Conference Carbon Footprint Report

Sample: Corporate Sustainability Summit

Sample Convention Hotel, Mumbai, Maharashtra, India · 2027-01-20 – 2027-01-21 · Report generated 08 November 2026, 09:19 UTC

Total footprint	Per participant	Per conference day	Data quality
111.78 t CO_2e	**223.6 kg CO_2e**	**55.89 t CO_2e**	**Medium (score 1.82)**
111,783 kg CO_2e (gross, location-based)	500 in-person participants	2 day(s)	Uncertainty: High

Gross vs. net. Gross emissions: **111.78 t CO_2e** (location-based electricity). Market-based total: 111.78 t CO_2e. Offsets/credits retired: 0.00 t CO_2e (reported separately; they do not reduce gross emissions). Gross minus offsets: 111.78 t CO_2e.



1. Conference information and reporting boundary

Item	Value
Conference	Sample: Corporate Sustainability Summit
Type / format	Corporate / Offline
Location	Sample Convention Hotel, Mumbai, Maharashtra, India
Dates / duration	2027-01-20 – 2027-01-21 (2 days)
Participants (in person)	440 attendees, 25 speakers, 25 organisers/staff, 10 volunteers = 500
Virtual attendees	0
Venue area	2,500 m²
Reporting boundary	Event-based operational boundary: delegate, speaker and staff travel; accommodation; venue energy; catering; waste; purchased event materials (GHG Protocol Scope 1, 2 and relevant Scope 3 categories 1, 3*, 5, 6, 7 for the organiser). *T&D;/WTT excluded.
Reporting period	2027-01-20 – 2027-01-21

2. Methodology

Emissions are calculated as **activity data × emission factor** for each activity, summed by category. The approach follows the GHG Protocol Corporate Standard, Scope 2 Guidance (dual reporting of location- and market-based electricity) and Scope 3 Standard, and ISO 14064-1 principles of relevance, completeness, consistency, accuracy and transparency. Results are expressed in CO_2-equivalent using IPCC AR5 100-year GWPs. Missing data is never treated as zero: categories without data are flagged 'not reported'. India-specific factors are prioritised (CEA grid factor, India GHG Program road and rail factors, CHSB India hotel benchmark); documented international proxies are used and flagged where no Indian factor is published.

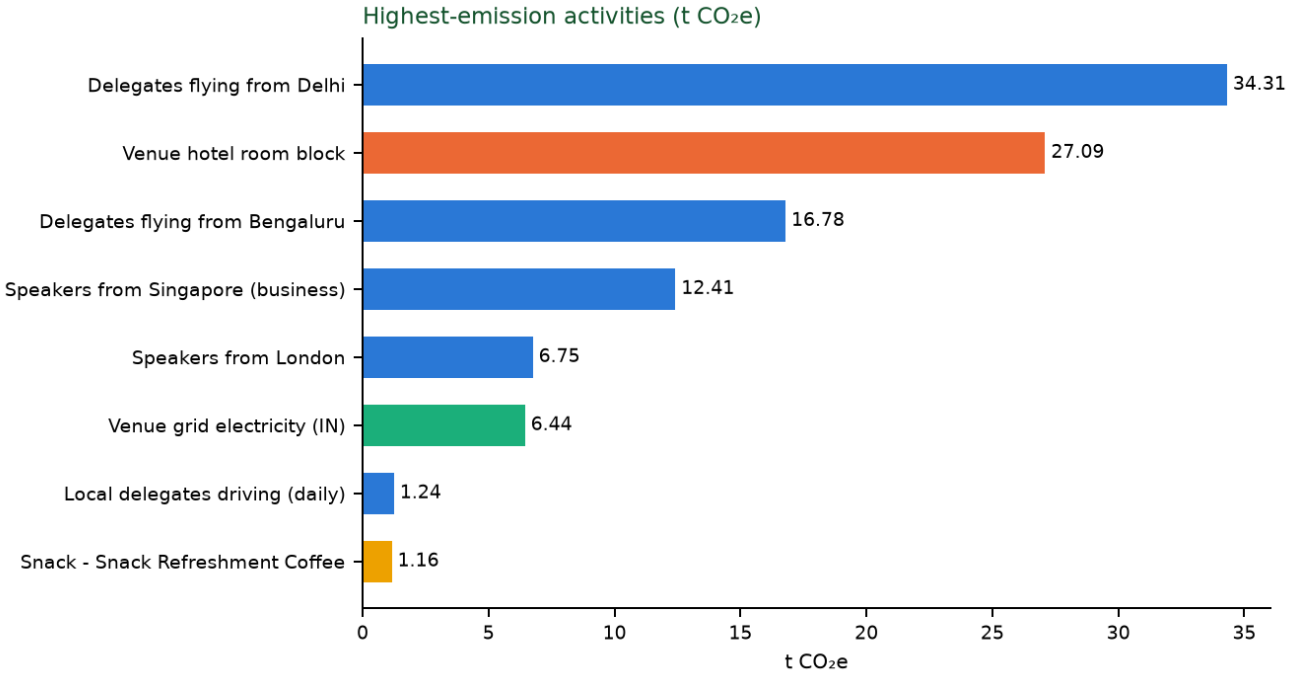
Category	Calculation	Primary factor source
Transportation	Public modes: passengers × km × legs × trips × kg/passenger-km. Vehicle modes: (passengers ÷ occupancy) × km × legs × trips × kg/vehicle-km. Flights by haul and cabin class, optional radiative forcing.	India GHG Program (2015) road & rail; DESNZ 2026 (aviation, coach, UK proxies)

Category	Calculation	Primary factor source
Accommodation	Room-nights × country benchmark (or hotel-specific factor, or actual hotel energy)	Cornell CHSB 2021 via DESNZ 2026
Venue energy	Metered kWh × grid factor (location-based); market-based nets certified renewables; fuels × combustion factors	CEA CO_2 Baseline v22.0 (FY2025-26); DESNZ 2026 fuels
Food & beverages	Meals × modelled per-meal factor, or kg ingredient × kg CO_2e/kg; bottles × PET mass × factor	Poore & Nemecek (2018) via OWID; ICMR-NIN My Plate (2024)
Waste	kg by material and disposal route × route factor (cut-off method)	DESNZ 2026 waste disposal
Materials	kg material (incl. pages → kg via ISO 216 × gsm) × cradle-to-gate factor; reusable items ÷ expected uses	DESNZ 2026 material use

3. Results by category

Category	Status	Emissions (kg CO_2e)	Emissions (t CO_2e)	Share
Transportation	calculated	72,099.9	72.100	64.5%
Accommodation	calculated	27,094.0	27.094	24.2%
Venue & energy	calculated	7,193.3	7.193	6.4%
Food & beverages	calculated	3,963.1	3.963	3.5%
Waste	calculated	215.5	0.216	0.2%
Materials & procurement	calculated	1,217.5	1.218	1.1%
Total (gross)		**111,783.3**	**111.783**	100%

Electricity: location-based 6,441.3 kg CO_2e; market-based 6,441.3 kg CO_2e. Per participant per day: 111.78 kg CO_2e.



4. Activity-level calculation table

Every reported number is traceable to its activity data and emission factor (factor id = key@dataset version).

#	Category / activity	Quantity	Factor	Source / factor id	kg CO_2e	Data
L0001	**Transportation** Delegates flying from Delhi 120 pax × 1,137 km × 2 leg(s) × 1 trip(s) × 0.12576	272,835.99 passenger.km	0.12576 kg CO2e/passenger.km	transport.flight.short_haul_uk.economy.with_rf@DESNZ-2026-v1.2	34,311.85	estimated high · proxy
L0002	**Transportation** Delegates flying from Bengaluru 80 pax × 834 km × 2 leg(s) × 1 trip(s) × 0.12576	133,466.20 passenger.km	0.12576 kg CO2e/passenger.km	transport.flight.short_haul_uk.economy.with_rf@DESNZ-2026-v1.2	16,784.71	estimated high · proxy
L0003	**Transportation** Speakers from Singapore (business) 5 pax × 3,922 km × 2 leg(s) × 1 trip(s) × 0.31656	39,218.22 passenger.km	0.31656 kg CO2e/passenger.km	transport.flight.international_non_uk.business.with_rf@DESNZ-2026-v1.2	12,414.92	estimated high · proxy
L0004	**Transportation** Speakers from London 4 pax × 7,213 km × 2 leg(s) × 1 trip(s) × 0.11704	57,703.28 passenger.km	0.11704 kg CO2e/passenger.km	transport.flight.long_haul_uk.economy.with_rf@DESNZ-2026-v1.2	6,753.59	estimated high
L0005	**Transportation** Delegates by train from Pune 60 pax × 190.0 km × 2 leg(s) × 1 trip(s) × 0.00793	22,800.00 passenger.km	0.0079293 kg CO2e/passenger.km	transport.rail.national.IN@IndiaGHGP-Rail-v1.0-2015	180.79	actual medium
L0006	**Transportation** Local delegates by suburban rail (daily) 100 pax × 20.0 km × 2 leg(s) × 2 trip(s) × 0.00807	8,000.00 passenger.km	0.00807 kg CO2e/passenger.km	transport.rail.suburban.IN@IndiaGHGP-Rail-v1.0-2015	64.56	actual medium
L0007	**Transportation** Airport taxis (209 pax ÷ 1.5 per vehicle) × 18.0 km × 2 leg(s) × 1 trip(s) × 0.06885	5,016.00 vehicle.km	0.068847 kg CO2e/vehicle.km	transport.car.small.cng.IN@IndiaGHGP-Road-v1.0-2015	345.34	actual medium

#	Category / activity	Quantity	Factor	Source / factor id	kg CO_2e	Data
L0008	**Transportation** Local delegates driving (daily) (80 pax ÷ 1.2 per vehicle) × 22.0 km × 2 leg(s) × 2 trip(s) × 0.21206	5,866.67 vehicle.km	0.21206 kg CO2e/vehicle.km	transport.car.suv.diesel.IN@IndiaGHGP-Road-v1.0-2015	1,244.11	actual medium
L0009	**Accommodation** Venue hotel room block 230.0 rooms × 2 nights × 58.90	460.00 room.night	58.9 kg CO2e/room.night	accommodation.hotel.india@DESNZ-2026-v1.2	27,094.00	actual medium
L0010	**Venue & energy** Venue grid electricity (IN) 9,500 kWh × 0.67804 (location-based)	9,500.00 kWh	0.67804 kg CO2e/kWh	electricity.grid.IN@CEA-v22.0	6,441.34	actual high
L0011	**Venue & energy** Diesel generator 150.0 litre × 2.66155	150.00 litre	2.6616 kg CO2e/litre	energy.fuel.diesel.litre@DESNZ-2026-v1.2	399.23	actual high
L0012	**Venue & energy** LPG 120.0 kg × 2.93936	120.00 kg	2939.4 kg CO2e/tonne	energy.fuel.lpg.tonne@DESNZ-2026-v1.2	352.72	actual high
L0013	**Food & beverages** Breakfast - Vegetarian 460 meals × 0.789	460.00 meal	0.7886 kg CO2e/meal	food.meal.vegetarian@MealModel-2026.1	362.76	actual medium
L0014	**Food & beverages** Lunch - Vegetarian 550 meals × 0.789	550.00 meal	0.7886 kg CO2e/meal	food.meal.vegetarian@MealModel-2026.1	433.73	actual medium
L0015	**Food & beverages** Lunch - Non Vegetarian Chicken 350 meals × 1.725	350.00 meal	1.7249 kg CO2e/meal	food.meal.non_vegetarian_chicken@MealModel-2026.1	603.72	actual medium
L0016	**Food & beverages** Lunch - Non Vegetarian Fish 100 meals × 2.101	100.00 meal	2.1009 kg CO2e/meal	food.meal.non_vegetarian_fish@MealModel-2026.1	210.09	actual medium
L0017	**Food & beverages** Dinner - Vegetarian 250 meals × 0.789	250.00 meal	0.7886 kg CO2e/meal	food.meal.vegetarian@MealModel-2026.1	197.15	actual medium
L0018	**Food & beverages** Dinner - Non Vegetarian Mutton 200 meals × 4.710	200.00 meal	4.7099 kg CO2e/meal	food.meal.non_vegetarian_mutton@MealModel-2026.1	941.98	actual medium
L0019	**Food & beverages** Snack - Snack Refreshment Coffee 2,000 meals × 0.578	2,000.00 meal	0.5778 kg CO2e/meal	food.meal.snack_refreshment_coffee@MealModel-2026.1	1,155.60	actual medium
L0020	**Food & beverages** PET bottles (material production) 1,500 bottles × 10 g = 15.00 kg × 3.862	15.00 kg	3.8616 kg CO2e/kg	materials.plastic.pet@DESNZ-2026-v1.2	57.92	actual medium
L0021	**Food & beverages** Water content (utility supply proxy) 0.750 m³ × 0.1913	0.75 m3	0.1913 kg CO2e/m3	water.supply@DESNZ-2026-v1.2	0.14	actual medium · proxy
L0022	**Waste** Food → landfill 200.0 kg × 0.70033	200.00 kg	0.70033 kg CO2e/kg	waste.food.landfill@DESNZ-2026-v1.2	140.07	actual medium · proxy
L0023	**Waste** Food → composting 60.0 kg × 0.00901	60.00 kg	0.0090069 kg CO2e/kg	waste.food.composting@DESNZ-2026-v1.2	0.54	actual medium · proxy
L0024	**Waste** Cardboard → recycling 70.0 kg × 0.00465	70.00 kg	0.0046536 kg CO2e/kg	waste.cardboard.recycling@DESNZ-2026-v1.2	0.33	actual medium · proxy
L0025	**Waste** Cardboard → unassigned (landfill assumed) 10.0 kg × 1.16451	10.00 kg	1.1645 kg CO2e/kg	waste.cardboard.landfill@DESNZ-2026-v1.2	11.65	default medium · proxy
L0026	**Waste** Plastic → landfill 40.0 kg × 0.00901	40.00 kg	0.0090069 kg CO2e/kg	waste.plastic.landfill@DESNZ-2026-v1.2	0.36	actual medium · proxy
L0027	**Waste** Plastic → recycling 20.0 kg × 0.00465	20.00 kg	0.0046536 kg CO2e/kg	waste.plastic.recycling@DESNZ-2026-v1.2	0.09	actual medium · proxy
L0028	**Waste** Mixed → landfill 120.0 kg × 0.52058	120.00 kg	0.52058 kg CO2e/kg	waste.mixed.landfill@DESNZ-2026-v1.2	62.47	actual medium · proxy
L0029	**Materials & procurement** Printed brochures (Virgin paper) 12,000 pages (double-sided) × 0.06237 m² × 130 g/m² = 48.65 kg × 1.344	48.65 kg	1.3436 kg CO2e/kg	materials.paper.virgin@DESNZ-2026-v1.2	65.36	actual medium · proxy

#	Category / activity	Quantity	Factor	Source / factor id	kg CO_2e	Data
L0030	**Materials & procurement** PVC banners & standees (PVC (banners, ID cards)) 40 kg = 40.00 kg × 2.942	40.00 kg	2.9424 kg CO2e/kg	materials.plastic.pvc@DESNZ-2026-v1.2	117.70	actual · medium · proxy
L0031	**Materials & procurement** Plastic ID cards (PVC (banners, ID cards)) 500 items × 5 g = 2.50 kg × 2.942	2.50 kg	2.9424 kg CO2e/kg	materials.plastic.pvc@DESNZ-2026-v1.2	7.36	actual · medium · proxy
L0032	**Materials & procurement** Polyester lanyards (PET (bottles; polyester polymer proxy)) 500 items × 12 g = 6.00 kg × 3.862	6.00 kg	3.8616 kg CO2e/kg	materials.plastic.pet@DESNZ-2026-v1.2	23.17	actual · medium · proxy
L0033	**Materials & procurement** Cotton conference bags (Textiles (cotton bags, fabric banners, T-shirts)) 500 items × 90 g = 45.00 kg × 22.310	45.00 kg	22.31 kg CO2e/kg	materials.textile.clothing@DESNZ-2026-v1.2	1,003.95	actual · medium · proxy

5. Emission factor sources and dataset versions

Factor id	Value	Source	Year / geography	Boundary
accommodation.hotel.india@DESNZ-2026-v1.2	58.9 kg CO2e/room.night	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	CHSB 2021 benchmark (held constant since 2022 DESNZ set) India	Hotel operational energy (Scope 1+2 of the hotel) allocated per occupied room-night; excludes food, laundry outsourcing and embodied emissions.
electricity.grid.IN@CEA-v22.0	0.67804 kg CO2e/kWh	CEA CO2 Baseline Database User Guide, Version 22.0	FY 2025-26 India (single national grid)	Weighted average CO2 emission of all grid-connected generation incl. renewables and captive injection, adjusted for cross-border transfers (net generation basis). Scope 2 location-based. Excludes T&D; losses and upstream fuel supply.
energy.fuel.diesel.litre@DESNZ-2026-v1.2	2.6616 kg CO2e/litre	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 Fuel property based (applicable internationally; UK fuel specification)	Scope 1 combustion emissions only; excludes well-to-tank.
energy.fuel.lpg.tonne@DESNZ-2026-v1.2	2939.4 kg CO2e/tonne	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 Fuel property based (applicable internationally; UK fuel specification)	Scope 1 combustion emissions only; excludes well-to-tank.
food.meal.non_vegetarian_chicken@MealModel-2026.1	1.7249 kg CO2e/meal	Modelled meal archetype: NIN 'My Plate for the Day' (2024) x Poore & Nemecek (2018)	2026 model India diet composition x global food factors	Farm-to-retail life cycle: land-use change, farm, animal feed, processing, transport, retail and packaging, plus food losses along the supply chain. Excludes consumer cooking and consumer-level waste.
food.meal.non_vegetarian_fish@MealModel-2026.1	2.1009 kg CO2e/meal	Modelled meal archetype: NIN 'My Plate for the Day' (2024) x Poore & Nemecek (2018)	2026 model India diet composition x global food factors	Farm-to-retail life cycle: land-use change, farm, animal feed, processing, transport, retail and packaging, plus food losses along the supply chain. Excludes consumer cooking and consumer-level waste.
food.meal.non_vegetarian_mutton@MealModel-2026.1	4.7099 kg CO2e/meal	Modelled meal archetype: NIN 'My Plate for the Day' (2024) x Poore & Nemecek (2018)	2026 model India diet composition x global food factors	Farm-to-retail life cycle: land-use change, farm, animal feed, processing, transport, retail and packaging, plus food losses along the supply chain. Excludes consumer cooking and consumer-level waste.
food.meal.snack_refreshment_coffee@MealModel-2026.1	0.5778 kg CO2e/meal	Modelled meal archetype: NIN 'My Plate for the Day' (2024) x Poore & Nemecek (2018)	2026 model India diet composition x global food factors	Farm-to-retail life cycle: land-use change, farm, animal feed, processing, transport, retail and packaging, plus food losses along the supply chain. Excludes consumer cooking and consumer-level waste.
food.meal.vegetarian@MealModel-2026.1	0.7886 kg CO2e/meal	Modelled meal archetype: NIN 'My Plate for the Day' (2024) x Poore & Nemecek (2018)	2026 model India diet composition x global food factors	Farm-to-retail life cycle: land-use change, farm, animal feed, processing, transport, retail and packaging, plus food losses along the supply chain. Excludes consumer cooking and consumer-level waste.
materials.paper.virgin@DESNZ-2026-v1.2	1.3436 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	Cradle-to-gate material production (extraction, processing, manufacture). Excludes use phase and end-of-life (end-of-life is reported under Waste).
materials.plastic.pet@DESNZ-2026-v1.2	3.8616 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	Cradle-to-gate material production (extraction, processing, manufacture). Excludes use phase and end-of-life (end-of-life is reported under Waste).
materials.plastic.pvc@DESNZ-2026-v1.2	2.9424 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	Cradle-to-gate material production (extraction, processing, manufacture). Excludes use phase and end-of-life (end-of-life is reported under Waste).
materials.textile.clothing@DESNZ-2026-v1.2	22.31 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	Cradle-to-gate material production (extraction, processing, manufacture). Excludes use phase and end-of-life (end-of-life is reported under Waste).
transport.car.small.cng.IN@IndiaGHGP-Road-v1.0-2015	0.068847 kg CO2e/vehicle.km	India Specific Road Transport Emission Factors, Version 1.0 (2015)	2015 India	Tank-to-wheel fuel combustion; includes 10% real-world uplift over type-approval CO2.

Factor id	Value	Source	Year / geography	Boundary
transport.car.suv.diesel.IN@IndiaGHGP-Road-v1.0-2015	0.21206 kg CO2e/vehicle.km	India Specific Road Transport Emission Factors, Version 1.0 (2015)	2015 India	Tank-to-wheel fuel combustion; includes 10% real-world uplift over type-approval CO2.
transport.flight.international_non_uk.business.with_rf@DESNZ-2026-v1.2	0.31656 kg CO2e/passenger.km	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 UK-registered/UK-route basis; non-UK international set from DfT/Innovata analysis	Direct fuel combustion (CO2, CH4, N2O). Includes 8% great-circle distance uplift. Includes radiative-forcing multiplier (1.7 on CO2) for non-CO2 climate effects.
transport.flight.long_haul_uk.economy.with_rf@DESNZ-2026-v1.2	0.11704 kg CO2e/passenger.km	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 UK-registered/UK-route basis; non-UK international set from DfT/Innovata analysis	Direct fuel combustion (CO2, CH4, N2O). Includes 8% great-circle distance uplift. Includes radiative-forcing multiplier (1.7 on CO2) for non-CO2 climate effects.
transport.flight.short_haul_uk.economy.with_rf@DESNZ-2026-v1.2	0.12576 kg CO2e/passenger.km	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 UK-registered/UK-route basis; non-UK international set from DfT/Innovata analysis	Direct fuel combustion (CO2, CH4, N2O). Includes 8% great-circle distance uplift. Includes radiative-forcing multiplier (1.7 on CO2) for non-CO2 climate effects.
transport.rail.national.IN@IndiaGHGP-Rail-v1.0-2015	0.0079293 kg CO2e/passenger.km	India Specific Rail Transport Emission Factors for Passenger Travel and Material Transport, Version 1.0 (2015)	FY 2014-15 India	Traction energy (diesel + electricity at then-current grid factor) allocated by passenger-km. No class (AC/sleeper) differentiation. Not for metro/MRTS.
transport.rail.suburban.IN@IndiaGHGP-Rail-v1.0-2015	0.00807 kg CO2e/passenger.km	India Specific Rail Transport Emission Factors for Passenger Travel and Material Transport, Version 1.0 (2015)	FY 2014-15 India	Traction energy (diesel + electricity at then-current grid factor) allocated by passenger-km. No class (AC/sleeper) differentiation. Not for metro/MRTS.
waste.cardboard.landfill@DESNZ-2026-v1.2	1.1645 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.
waste.cardboard.recycling@DESNZ-2026-v1.2	0.0046536 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.
waste.food.composting@DESNZ-2026-v1.2	0.0090069 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.
waste.food.landfill@DESNZ-2026-v1.2	0.70033 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.
waste.mixed.landfill@DESNZ-2026-v1.2	0.52058 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.
waste.plastic.landfill@DESNZ-2026-v1.2	0.0090069 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.

Factor id	Value	Source	Year / geography	Boundary
waste.plastic.recycling@DESNZ-2026-v1.2	0.0046536 kg CO2e/kg	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	GHG Protocol Scope 3 Cat.5 'cut-off' approach: recycling/combustion/composting carry only collection and transport to the treatment site (benefits of recycling or energy recovery are not credited). Landfill includes methane from decomposition with UK landfill-gas capture rates.
water.supply@DESNZ-2026-v1.2	0.1913 kg CO2e/m3	UK Government GHG Conversion Factors for Company Reporting 2026 - flat file (revised, v1.2)	2026 United Kingdom (proxy)	Utility treatment and distribution energy.

Dataset versions used: CEA-v22.0, DESNZ-2026-v1.2, IndiaGHGP-Rail-v1.0-2015, IndiaGHGP-Road-v1.0-2015, MealModel-2026.1. Factor set FS-2026.1, engine 1.0.0, calculated 2026-11-08T09:19:28+00:00.

6. Assumptions and exclusions

Assumptions

• GWP basis: IPCC AR5 100-year GWPs (CH4 = 28, N2O = 265), consistent with DESNZ 2026.

• Electricity is reported location-based (headline) and market-based (renewable certificates at zero, remainder at grid average because no Indian residual-mix factor is published).

• Flight distances are great-circle distances; DESNZ factors already include an 8% routing uplift, so no further uplift is applied.

• Flights not touching the UK use DESNZ short-haul factors up to 3,700 km and DESNZ 'International, to/from non-UK' factors above 3,700 km.

• Vehicle modes (car, taxi, motorcycle, auto-rickshaw, EV) are calculated per vehicle-km: vehicles = passengers / occupancy unless the number of vehicles is given; public modes (air, rail, bus) per passenger-km.

• India-specific road and rail factors (India GHG Program 2015) are published as CO2 only and are converted to CO2e using the CH4/N2O share of the analogous DESNZ 2026 factor.

• India grid factor (CEA v22.0, CO2 only) is converted to CO2e using IPCC 2006 default CH4/N2O factors for coal combustion (+0.45%).

• Hotel emissions use the Cornell CHSB 2021 country median per occupied room-night (via DESNZ); no hotel-class differentiation is applied because no published class-specific factors exist.

• Food factors are Poore & Nemecek (2018) global means (farm-to-retail). Modelled meals use the ICMR-NIN 'My Plate for the Day' (2024) composition; protein-dish portion (100 g raw) is a modelling assumption.

• Waste factors follow the DESNZ / GHG Protocol cut-off method: recycling, composting and combustion carry collection and transport only; no avoided-emission credits are claimed.

Entry-specific notes

• Bottle production only; filling, transport and refrigeration excluded.

• Converted from kg CO2e/tonne.

• DESNZ UK-route factor applied as international proxy (no India-specific aviation factor is published).

• Disposal route not specified - landfill assumed (conservative).

• Great-circle distance BLR–BOM estimated from airport coordinates (OurAirports).

• Great-circle distance DEL–BOM estimated from airport coordinates (OurAirports).

• Great-circle distance LHR–BOM estimated from airport coordinates (OurAirports).

• Great-circle distance SIN–BOM estimated from airport coordinates (OurAirports).

• Location-based Scope 2 (GHG Protocol Scope 2 Guidance).

• Only a small-car CNG factor is published for India; applied to this vehicle.

• Paper mass from ISO 216 sheet area × grammage; printing energy/ink excluded.

• Per-meal factor is a modelled archetype (NIN plate × Poore & Nemecek); use ingredient quantities for higher accuracy.

• Scope 1 direct combustion.

• Taxi calculated with the car factor for its size/fuel; empty running between fares is excluded.

• UK landfill gas-capture assumptions - likely an UNDER-estimate for Indian landfills/dumpsites.

• UK water-supply factor used as proxy.

Exclusions

• Non-CO2 effects of aviation are included only when the radiative-forcing option is enabled.

• Well-to-tank (upstream fuel and electricity supply) emissions and electricity T&D; losses.

• Emissions from virtual participation (devices, networks and data centres) - no robust, openly published factor was found.

• Venue construction, furniture and capital goods (GHG Protocol Scope 3 Category 2).

• Consumer-stage cooking energy beyond what is captured in venue electricity/fuel data.

• Printing process energy and inks (paper material only is counted).

• Biogenic CO2 from waste treatment (reported as zero, consistent with DESNZ).

7. Actual vs. estimated data, data quality and uncertainty

Activity data type	Emissions (kg CO_2e)	Share	Lines
Actual	41,506.6	37.1%	28
Estimated	70,265.1	62.9%	4
Default	11.6	0.0%	1

Emissions-weighted factor data-quality score: **1.82** (medium; 1 (best) – 5 (worst), emissions-weighted pedigree score of the factors used). Share of emissions calculated with international proxy factors: 58.1%. Qualitative uncertainty rating: **High**. Scores follow a pedigree approach (geographical, temporal and technological representativeness and source reliability, 1 = best, 5 = worst). Most sources do not publish statistical uncertainty ranges, so no numerical confidence interval is claimed.

8. Sustainability recommendations

Recommendations are generated from this conference's emission breakdown and quantified with the scenario engine using the same factors.

1. Cut long-distance air travel (Transportation) — estimated reduction up to **25.20 t CO_2e (22.5%, location-based (gross))** [Hybrid format: share of attendees online]

Why: Air travel is 62.9% of this conference's footprint.

• Invite regional speakers or offer remote keynote slots for distant speakers.

• Offer a hybrid/virtual registration tier for international delegates.

• Encourage rail for journeys under ~1,000 km and economy class for necessary flights.

• Choose a venue that minimises total delegate travel distance.

2. Reduce and decarbonise venue electricity (Venue & energy) — estimated reduction up to **6.44 t CO_2e (5.8%, market-based)** [Renewable electricity]

Why: Grid electricity is 5.8% of the footprint (location-based).

• Procure renewable electricity (green tariff / I-RECs) - reported in the market-based total.

• Choose venues with on-site solar or certified energy management (e.g. ISO 50001).

• Set AC to 24–26 °C (BEE guidance), switch to LED lighting and power down equipment between sessions.

3. Go digital and reusable (Materials & procurement) — estimated reduction up to **0.92 t CO_2e (0.8%, location-based (gross))** [Reusable banners, lanyards and kits]

Why: Materials & procurement are 1.1% of the footprint.

• Use a conference app/website instead of printed programmes.

• Use undated, reusable banners and fabric signage; collect lanyards and badge holders for reuse.

• Avoid giveaway kits or choose recycled-content, locally made items.

4. Choose lower-carbon accommodation (Accommodation)

Why: Accommodation is 24.2% of the footprint (default CHSB country benchmark used unless hotel data supplied). Savings depend on the chosen hotel's own data, so no generic reduction is estimated.

• Block rooms in hotels within walking distance of the venue to also cut local transport.

• Prefer hotels that report energy data or hold recognised green certifications; request their kg CO2e per room-night.

• Encourage room sharing and avoid unnecessary extra nights through programme scheduling.

5. Improve the data behind this footprint (Data quality)

Why: 62.9% of emissions rely on estimated or default activity data.

• Collect actual travel data (origin, mode, class) through the registration form.

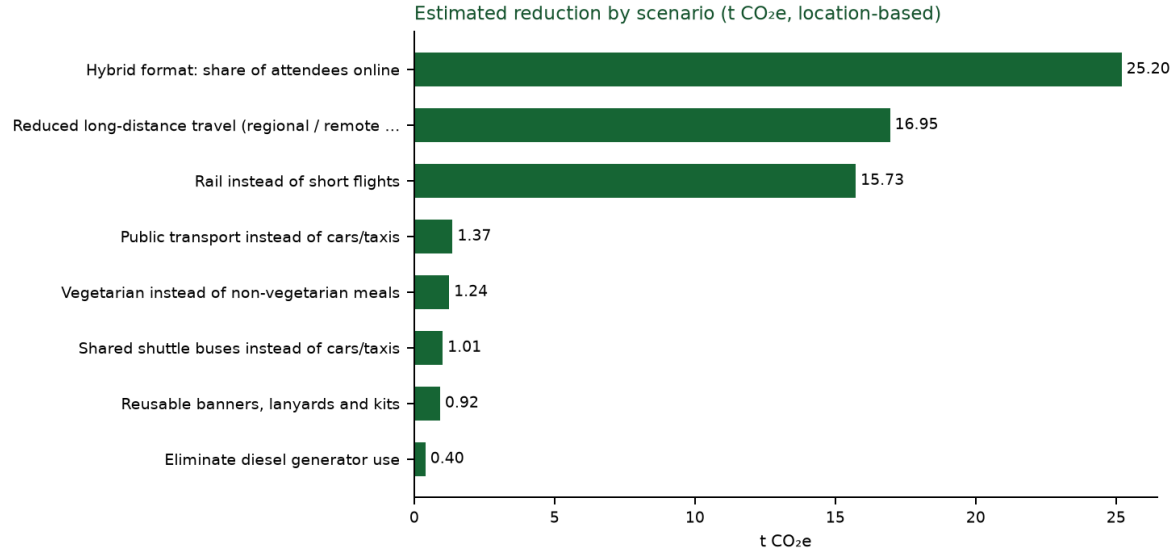
• Obtain metered electricity, generator fuel logs and hotel energy data.

• Weigh waste by stream with the venue/waste contractor.

9. Scenario comparison (what-if analysis)

Scenario	Baseline (t)	Scenario (t)	Reduction (t)	Reduction	Description
Shared shuttle buses instead of cars/taxis	111.78	110.77	1.01	0.9%	289 road travellers moved from cars/taxis to shared coaches (100%).
Public transport instead of cars/taxis	111.78	110.42	1.37	1.2%	289 car/taxi travellers switched to rail (>50 km) or city bus (100%).
Rail instead of short flights	111.78	96.06	15.73	14.1%	80 flight passengers on routes ≤ 1000 km switched to rail (rail distance taken equal to flight great-circle distance - real rail routes are usually longer).
Reduced long-distance travel (regional / remote speakers)	111.78	94.84	16.95	15.2%	25% fewer journeys by flight (e.g. regional speakers, remote talks).

Scenario	Baseline (t)	Scenario (t)	Reduction (t)	Reduction	Description
Hybrid format: share of attendees online	111.78	86.59	25.20	22.5%	132 attendees (30%) participate online. Travel, accommodation and catering scale down for attendee groups; venue energy, waste and materials unchanged; virtual-participation emissions not quantified.
Vegetarian instead of non-vegetarian meals	111.78	110.54	1.24	1.1%	650 non-vegetarian meals (100%) replaced with vegetarian meals.
Renewable electricity	111.78	111.78	0.00	0.0% market: 5.8%	9,500 kWh (100%) covered by renewable energy certificates / green tariff. This lowers the MARKET-based total only; the location-based (headline) total is unchanged by design.
Eliminate diesel generator use	111.78	111.38	0.40	0.4%	Diesel generator use (150 L) eliminated through a reliable grid connection. The replacement grid electricity is NOT added (generator kWh output unknown) - result is an upper bound of savings.
Reusable banners, lanyards and kits	111.78	110.86	0.92	0.8%	4 single-use items (banners, lanyards, badges, bags, kits) made reusable over 5 events.
Digital instead of printed materials	111.78	111.73	0.05	0.1%	Printed paper reduced by 80% via digital programmes/apps (device energy not counted).
Reduced food waste	111.78	111.71	0.07	0.1%	Food waste reduced by 50% through better catering planning. Only disposal emissions are modelled; avoided food production would add further savings.
Compost food waste	111.78	111.65	0.14	0.1%	All food waste segregated and composted instead of landfilled.



10. Reporting statement

This report was generated on 08 November 2026, 09:19 UTC from calculation #10. The stored calculation retains the exact input data and the identifier and value of every emission factor used, so the results are reproducible even after the factor database is updated. This is a self-declared, unverified estimate; third-party verification (e.g. to ISO 14064-3) is recommended before public claims. Offsets, if any, are reported separately and are not deducted from gross emissions.